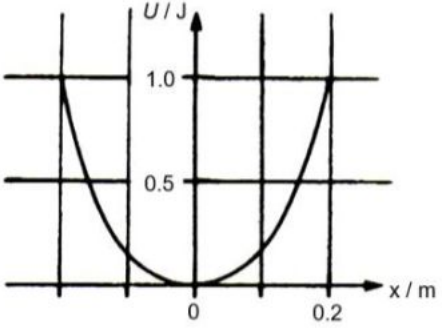
10 A particle of mass 4.0 kg moves with simple harmonic motion and its potential energy U varies with position x as shown in the figure below.



What is the period of oscillation of the particle?

A 0.25 s **B** 0.89 s **C** 1.8 s **D** 2.5 s

Ans: C

$$E = \frac{1}{2} m\omega^2 x_o^{\ 2}$$

$$1.0 = \frac{1}{2}(4.0)(\frac{2\pi}{T})^2(0.2)^2$$

$$T = 1.78 \text{ s}$$